

Distributed Manufacturing of an ETH Zürich Oxygen Sensor

A complete four-phase production roadmap, derived from publicly available supplementary data

Source paper: Wettstein et al., Advanced Science (2025), DOI 10.1002/advs.202405694

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Foreword: What This Document Demonstrates

Atom Factory is the manufacturing layer of ttOS. It uses the Contract Logic Engine (CLE) to engage university labs at scale through standardised consulting and services agreements, turning the global academic research infrastructure into a distributed production network for breakthrough materials.

This document is a **worked example** of the Atom Factory method, applied to a single published invention: the SWCNT–TiO₂–Re dye-sensitized oxygen sensor developed at ETH Zürich and published by Wettstein et al. in Advanced Science (2025). The complete roadmap that follows — from the original supplementary data, through a four-phase manufacturing plan, to a fully costed production model at 100,000 sensors per year — was constructed entirely from publicly available sources: the original paper, its supplementary information, the ETH Transfer technology sheet, and standard market data on materials, labour, and equipment.

It demonstrates what becomes possible when CLE-enabled contracts make it economical to commission production runs across twenty universities in parallel, when central CRO synthesis replaces twenty redundant in-lab batches, and when fabrication concentrates at low-cost nodes once the synthesis bottleneck is removed.

A note on status. This roadmap has not yet been executed. No sensors have been manufactured under the programme described here, no CRO contract has been placed, no university partner has been engaged. The document is a method demonstration, not a record of a delivered project. Atom Factory is currently seeking foundation funding and university partners to begin Phase 1 in 2026. The purpose of publishing this roadmap is to show, in full and defensible detail, what a real Atom Factory programme would look like — and to invite the conversations that would make it real.

The substance of the roadmap that follows is unchanged from the internal working version: every cost figure, every supplier name, every university capability assessment, and every risk has been carried over without simplification. What has been added is the framing above. What has been removed is nothing.

Executive Summary

This document tells the complete story of how a single university research paper — a supplementary data file from ETH Zürich describing a novel dye-sensitized oxygen chemiresistor — was systematically transformed into a three-phase global manufacturing roadmap.

The sensor in question is a remarkable device: a thin-film stack of single-walled carbon nanotubes (SWCNTs), amorphous titanium dioxide (TiO_2), and a rhenium carbonyl bipyridine sensitizer dye deposited on a gold interdigitated electrode. Under green visible light, it detects oxygen at concentrations as low as 157 parts per billion, is selective against all common interfering gases, tolerates humidity from 0 to 80%, and remains stable for over three months. It is, in short, a publishable result with real commercial potential.

The process that follows began with a careful read of the supplementary data: extracting which materials were commercially available and which had to be synthesised, understanding the synthesis bottleneck, mapping which universities worldwide could realistically execute the full process, and then building a rigorous cost model that distinguished university overhead, regional labour rates, material scale economies, and supply-chain risks.

The result is a three-phase roadmap:

Phase 1 ~\$120k <i>100 sensors, 20 universities</i>	Phase 2 ~\$42k <i>CRO synthesis, 100g Re dye</i>	Phase 3 (pilot) ~\$370k <i>5,000 sensors, India + Korea</i>
Phase 3 per sensor ~\$74 <i>vs ~\$1,200 at uni lab scale</i>	Electrode risk Critical <i>50% of Phase 3 cost</i>	Re supply risk Watch <i>\$6,112/kg and rising</i>

If Phase 3 pilot validation confirms use-case performance, the roadmap extends to full-scale production at 100,000 sensors per year, bringing the per-sensor cost to approximately \$14–\$20 depending on electrode sourcing.

1. Starting Point: The Supplementary Data

1.1 What the Paper Describes

The source paper, “A Dye-Sensitized Sensor for Oxygen Detection under Visible Light” (Wettstein et al., Adv. Sci. DOI 10.1002/advs.202405694), describes a three-component chemiresistor fabricated by depositing a composite film of SWCNTs and amorphous TiO_2 onto gold interdigitated electrodes, followed by sensitisation with a phosphonate-anchored rhenium carbonyl dye. The sensor responds to oxygen under illumination by green LED, with the excited-state rhenium complex acting as a photoredox mediator to shuttle charge between oxygen and the SWCNT network.

The supplementary information provided a complete synthetic and fabrication protocol, detailed characterisation data, and full sensing measurement protocols. This is the starting document for everything that follows.

1.2 Extracting the Input Materials

The first analytical task was to read the supplementary data and produce a clean inventory of every material required, categorised by whether it was commercially available off the shelf or required in-house synthesis or purification. The results are summarised in the table below.

Material
Commercially Available
Purification Required
Supplier
Ti(OiPr)₄ 99.999%

Yes

No — use as received

Sigma-Aldrich 377996

[Re(CO)₅Br]

Yes

No — use as received

Sigma-Aldrich 327638

P3-SWNT (>90% purity)

Yes

No — dry under vacuum only

Carbon Solutions Inc.

4,4'-Dibromo-2,2'-bipyridine

Yes

No

Chem-Impex / abcr

PPh₃ 99%

Yes

No

Sigma-Aldrich

Pd(PPh₃)₄ 99%

Yes

No

Sigma-Aldrich 216666

Diethyl phosphite

Yes

No

Sigma-Aldrich

Bromotrimethylsilane 97%

Yes

No (moisture-sensitive)

Sigma-Aldrich 194409

Et₃N

Yes

No

Sigma-Aldrich

Anhydrous DMSO

Yes

No — use Sure/Seal

Sigma-Aldrich 276855

Anhydrous toluene

Yes

No — use Sure/Seal

Sigma-Aldrich 244511

Anhydrous CHCl₃, MeOH, EtOH

Yes

No — use Sure/Seal

Sigma-Aldrich

IDEAU200 gold IDE electrodes

Yes

No

Metrohm DropSens DRP-IDEAU200-U50

N3 dye (control only)
 Yes
 Yes — recrystallisation
 Solaronix / Ossila
 N719 dye (control only)
 Yes
 Yes — recrystallisation
 Solaronix / Ossila
 [(Pbpy)(CO)₃ReBr] Re sensitizer
 NO
 N/A — must be synthesised
 No commercial source exists

1.3 The Synthesis Bottleneck

Critical finding: [(Pbpy)(CO)₃ReBr] has no commercial source anywhere in the world. Every sensor requires this compound to be synthesised from scratch. This is the single most important constraint in the entire manufacturing chain.

Every other input can be purchased from a catalogue and used as received. The rhenium sensitizer dye — the compound that gives the sensor its visible-light response — does not exist commercially. It must be made in a two-step synthesis starting from [Re(CO)₅Br] and 4,4'-dibromo-2,2'-bipyridine.

This synthesis has three significant requirements that immediately constrain which facilities can perform it:

- An argon-filled glovebox (or at minimum a well-maintained Schlenk line) is required for the [Re(CO)₅Br] complexation step, which is conducted under strictly inert atmosphere conditions.
- A Pd-catalysed phosphonate coupling (Pd(PPh₃)₄, 110°C reflux in toluene, 15 hours under argon) is required for the first step, demanding competent Schlenk or glovebox technique.
- Purification requires flash chromatography (silica, 0–4% MeOH in DCM gradient), followed by a silyl deprotection step, a precipitation sequence, and high-vacuum drying.

The overall yield of the two-step sequence is approximately 29% (44% for Step 1, 65% for Step 2). This yield factor directly sets the amount of starting material that must be procured for any given production target.

The Two-Step Synthesis in Brief

Step
 Reaction
 Conditions
 Yield

Step 1

Pd-catalysed phosphonate coupling of dibromo-bipyridine

Pd(PPh₃)₄, Et₃N, toluene, 110°C, 15h, argon

~44%

Step 2a

Complexation with [Re(CO)₅Br]

1:1, dry toluene, 80°C, 19h (microwave or oil bath)

~94%

Step 2b

Silyl deprotection with bromotrimethylsilane

TMSBr (68 eq), CHCl₃, 65°C, 24h, then MeOH

~65%

Overall

4,4'-dibromo-bpy → [(Pbpy)(CO)₃ReBr]

Two steps, one purification

~29%

Confirmation of successful synthesis requires ¹H NMR (DMSO-d₆: δ 9.17, 8.75, 7.92 ppm), ³¹P NMR (δ 5.56 ppm), FT-IR (CO bands at 2033 and 1915 cm⁻¹), and UV-vis (DMF: λ_{max} 301 nm, λ_{MLCT} 389 nm). This characterisation is non-negotiable: without it, the active dye cannot be confirmed and sensor performance will be unpredictable.

2. Who Can Make This? The University Capability Map

2.1 The Two-Tier Problem

The synthesis bottleneck immediately creates a two-tier manufacturing challenge. Batch 1 — the Re sensitizer synthesis — requires a fully equipped inorganic/organometallic chemistry laboratory: glovebox or Schlenk line, microwave or high-temperature reflux capability, flash chromatography, high vacuum, and NMR access. Batch 2 — electrode fabrication — requires no glovebox at all; it needs only an airbrush, a heating plate, a syringe pump, an ultrasonic bath, and basic electrical testing equipment.

This split means that Tier 1 institutions (full synthesis capability) are required for Phase 1, while Tier 2 and Tier 3 institutions can handle Phase 3 fabrication once centralised Re sensitizer supply is established.

**Tier
Required Capability
Suitable For**

Tier 1

Glovebox or Schlenk, Pd catalysis, flash chromatography, NMR, FT-IR

Phase 1 (Batch 1 + Batch 2)

Tier 2

Airbrush, heating plate, syringe pump, ultrasonic bath, resistance meter

Phase 3 (Batch 2 only)

Tier 3

Basic wet lab and electrical testing

Phase 3 QC and assembly only

2.2 Verification Strategy

There is no global database that catalogues every piece of equipment at every university chemistry department. However, a practical three-tier verification strategy was developed that can confirm the full list of ~80 qualifying institutions within two to three working days.

For approximately 60 institutions — all G20-country research universities with confirmed organometallic publication output in the last three years — the assumption of full Tier 1 capability is well-founded and requires no further verification. If a group has published a paper in the last three years involving air-sensitive organometallic synthesis, they have the equipment. For a further 15 institutions where the primary research focus is materials science rather than organometallic chemistry, a ten-minute Google Scholar search confirms whether a relevant group exists. For the remaining 5 institutions (primarily newer South American or Central/Eastern European universities), a single email to the departmental facilities coordinator resolves the question definitively.

2.3 The Global Tier 1 University List

The
following
institutions
were
identified as
having
confirmed or

near-certain
Tier 1
capability for
Phase 1
participation.
Approximate
ly 80
institutions
qualify
globally; 20
will be
selected for
Phase 1.

Region

Institutions (selected examples)

Count

USA

MIT, Caltech, UC Berkeley, Stanford, Harvard, UIUC, Northwestern, UCLA, Georgia Tech, Columbia, UChicago, Yale, Princeton, U Michigan, Johns Hopkins

21

Canada

U Toronto, McGill, UBC, U Alberta, McMaster, U Waterloo

6

UK

Oxford, Cambridge, Imperial, UCL, Manchester, Edinburgh, Bristol, York, Warwick, Birmingham, Nottingham, Liverpool, St Andrews, Glasgow

14

Switzerland

ETH Zürich, EPFL, U Basel, U Zurich

4

Germany

LMU Munich, TU Munich, Heidelberg, RWTH Aachen, Würzburg, Münster, TU Dresden, Göttingen, FAU Erlangen, Freiburg, Mainz, Ruhr Bochum

14

France / Benelux / Scandinavia

Paris-Saclay, Sorbonne, Strasbourg; KU Leuven, Ghent; TU Delft, Groningen, Utrecht; Uppsala, Stockholm, Lund, Copenhagen, DTU, Aarhus

16

India

IIT Bombay, IIT Madras, IIT Delhi, IIT Kanpur, IIT Kharagpur, IISc Bangalore, IISER Pune

7

South Korea

KAIST, Seoul National University, POSTECH, Yonsei, SKKU

5

Singapore / Australia

NUS, NTU; ANU, U Melbourne, U Sydney, U Queensland

6

South America / CEE

USP, UNICAMP, UBA Argentina; IOCB Prague, Polish Academy of Sciences

5

3. The Cost Model: Labour Rates and Charge Rates

3.1 Regional Labour Rates

University production costs are dominated by labour. The effective hourly rate for a postdoctoral researcher — the appropriate skill level for this synthesis and fabrication work — varies by approximately 15-fold between the most expensive and least expensive regions in the qualified university list. The following rates include a 20–30% facilities and administrative (F&A)

overhead
applied to
labour,
consistent with
typical
university
research
project
accounting.

Region

Postdoc Annual Salary

Effective Hourly Rate (incl. OH)

Cost Tier

United States

~\$61,000/yr (NIH NRSA minimum)

~\$37/hr

High

Switzerland (ETH/EPFL)

~CHF 92,500/yr (~\$99,800 USD)

~\$60/hr

Highest

Germany

~€59,000/yr (~\$64,000 USD)

~\$40/hr

High

United Kingdom

~£34,000/yr (~\$43,000 USD)

~\$26/hr

Mid

South Korea

~₩50,600,000/yr (~\$37,000 USD)

~\$23/hr

Mid-Low

India (IIT/IISc)

~₹660,000/yr (~\$7,900 USD)

~\$5.70/hr

Low

3.2 Equipment and Facility Charge Rates

In addition to labour, core facility instruments carry their own charge rates. The most significant for this production process are NMR spectrometers, which are required to confirm successful Re sensitizer synthesis. The following rates represent typical internal academic user pricing.

Instrument

Internal Academic Rate

Relevance to This Process

NMR 400–600 MHz (solution)

\$10–\$25/hr

Required: ^1H + ^{31}P confirmation of Re dye

FT-IR spectrometer

~\$5–15/run (consumables only)

Required: CO bands 2033 and 1915 cm^{-1}

UV-vis spectrophotometer

~\$5–10/run

Required: λMLCT 389 nm confirmation

Flash chromatography (Biotage)

Included in OH or ~\$20–50/run

Required: Step 1 purification

High vacuum pump

Included in lab overhead

Required: product drying between steps

Airbrush + heating plate

~\$80–\$400 capital (amortised)

Batch 2 spray-coating of electrodes

Resistance meter / DAQ

~\$0.50/electrode

QC check: target 1–5 k Ω per device

4. Phase 1 — Distributed University Pilot: 100 Sensors across 20 Universities

4.1 Design of the Pilot

Phase 1 serves three purposes: it validates that the ETH Zürich protocol transfers faithfully to other institutions; it generates 100 sensors for use-case testing and application validation; and it begins building the international network of fabrication expertise that will underpin Phase 3.

The design calls for 20 universities — selected from the 80 in the qualified list — to each perform the complete synthesis and fabrication process independently. Each site produces 5 finished sensors (plus one spare to absorb yield loss) from a single Batch 1 synthesis run and a single Batch 2 fabrication run.

The key inefficiency of Phase 1 is deliberate: each site running its own Batch 1 synthesis generates redundant overhead but maximises protocol learning and independence. This is correct for a validation phase.

4.2 Sensor Production per Site

- One complete Batch 1 Re sensitizer synthesis (6 person-days: two multi-day reflux steps, one column chromatography, one silyl deprotection, high-vacuum drying, NMR/IR/UV-vis confirmation)
- One Batch 2 fabrication run producing 6 SWCNT-TiO₂-Re electrodes (2 person-days: SWCNT-TiO₂ preparation, 4-week aging in water, spray-coating, overnight drying, sensitizer soaking, vacuum drying)

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| • | • | • | • |

QC:
resistance
testing all 6
electrodes,
targeting 1–5
kΩ; pass
threshold
~80%, giving
5 shippable
sensors per
site

4.3 Cost by Region

Region Group

Universities

Cost per Site (5 sensors)

Subtotal

US / Switzerland (Tier 1 highest)

8

\$4,200–\$4,800

~\$35,200

Germany / UK / Netherlands (Tier 1 mid)

6

\$2,800–\$3,200

~\$18,000

South Korea — KAIST, SNU, POSTECH (Tier 1 mid-low)

3

\$2,200–\$2,600

~\$7,200

India — IIT Bombay, IIT Madras, IISc (Tier 1 low)

3

\$1,400–\$1,800

~\$4,800

TOTAL — 20 universities

20

blended ~\$3,400

~\$68,000

4.4 Phase 1 Per-University Cost Breakdown (Mid Estimate)

Cost Item

Amount

Notes

Batch 1 materials (Re synthesis reagents)

~\$708

Fixed regardless of output quantity

Batch 2 materials (6 IDEAU200 electrodes + chemicals)

~\$132

\$10/electrode × 6 + solvents, consumables

Synthesis labour (6 person-days)

\$380–\$1,840

Varies by regional rate

Fabrication labour (0.4 person-days)

\$25–\$120

Varies by regional rate

NMR + FT-IR + UV-vis confirmation

~\$80

Internal academic facility rates

University F&A overhead (30% on labour)

\$120–\$590

Applied to all labour

Consumables and contingency

~\$50

Glassware, filters, Ar gas usage

Per-sensor cost at Phase 1: ~\$680. This is high because a full Batch 1 synthesis (6 person-days, \$708 in reagents) is amortised over only 5 sensors per site. Eliminating this redundancy is the entire rationale for Phase 2.

4.5 Phase 1 Deliverables and Exit Criteria

- 100 confirmed sensors (resistance 1–5 kΩ, UV-vis λMLCT 389 nm confirmed) distributed to use-case testing partners
- Protocol transfer confirmed at ≥15 of 20 sites (i.e. ≥75% yield on Batch 1 confirmation by NMR/IR)
- First use-case data: response to target oxygen concentrations under green LED, selectivity vs. interfering gases, humidity tolerance
- Identification of 3–5 low-cost fabrication nodes (India, Korea) for Phase 3 scale-up

- Recommendation on OEM electrode substitution based on parallel validation experiment (200 test electrodes from Chinese OEM supplier)

5. Phase 2 — CRO Bulk Synthesis of the Re Sensitizer

5.1 The Strategic Logic

Phase 1 reveals the fundamental economics problem of distributed synthesis: each of 20 universities runs an independent full Batch 1 synthesis, each burning 6 person-days of skilled organometallic chemist time and \$708 in reagents to produce enough Re sensitizer for 5 sensors. The total Phase 1 labour cost for synthesis alone, across all 20 sites, is approximately \$30,000–\$40,000. This is inherently inefficient.

Phase 2 solves this by outsourcing all Batch 1 synthesis to a single specialist contract research organisation (CRO). One well-resourced synthetic chemistry CRO, running the two-step process at 100 gram scale, produces enough $[(Pbpy)(CO)_3ReBr]$ for approximately 2 million sensor soaking doses — more than sufficient for Phase 3 at 5,000 sensors, Phase 4 at 100,000 sensors, and significant reserve. The 20 parallel university synthesis runs, each with their own overhead and variability, are replaced by a single quality-controlled batch with full analytical certification.

5.2 CRO Cost Model — 100g Final Product

Item

Quantity / Basis

Cost Estimate

$[Re(CO)_5Br]$ precursor (140g required at 29% yield)

Bulk procurement, ~\$90/g at 100g+ volume

~\$12,600

4,4'-Dibromo-2,2'-bipyridine (~100g batch)

Bulk pricing ~\$80/g

~\$8,000

$Pd(PPh_3)_4$ catalyst (gram-scale amount)

Bulk price

~\$2,500

PPh_3 , diethyl phosphite, bromotrimethylsilane, Et_3N

Bulk solvents and reagents

~\$600

Anhydrous solvents: toluene, CHCl₃, MeOH (bulk drums)

Industrial grade, pre-dried

~\$500

CRO labour (~3 chemist-weeks, organometallic specialty)

FTE rate at European CRO

~\$9,000–\$14,000

Purification (flash chromatography, precipitation, QC)

NMR, IR, UV-vis, HPLC certification

~\$2,500

Argon-sealed packaging + cold-chain shipping to nodes

Multi-site distribution

~\$500

TOTAL CRO INVOICE

~\$36,000–\$46,000

At approximately \$420/g, the CRO route is 65% cheaper than university-scale production at

~\$1,200/g. More importantly, it delivers a homogeneous, fully characterised batch with CoA (certificate of analysis), eliminating the inter-site variability that would otherwise affect Phase 3 yield and performance.

5.3 Recommended CRO Partners

CRO

Location

Relevant Capability

Recommendation

Taros Discovery

Germany (+ Asia pilot plant)

Organometallic, inert atmosphere, pilot scale

Priority recommendation

Rieke Metals

USA

40+ years air-sensitive organometallic, inert atmosphere standard

Priority recommendation

ChiroBlock

Germany

Fixed-fee-for-service guaranteed delivery model

Good for fixed budget

Enamine

EU / Ukraine
Large FTE capacity, lower cost, broad chemistry
Verify Re capability first
Life Chemicals
EU
Materials science track record
Confirm organometallic depth

5.4 Critical Supply Risk: Rhenium Metal

WARNING: Rhenium metal price has risen +206% since January 2024 and +146% in 2025 alone, reaching \$6,112/kg in March 2026. It is now classified as a US critical mineral. A fixed-price CRO contract should be placed immediately.

Despite this dramatic price spike, the rhenium metal content per finished sensor is only approximately 0.4mg — a metal cost of approximately \$0.002 per sensor, which is negligible. The real financial risk is the $[\text{Re}(\text{CO})_5\text{Br}]$ precursor cost for the Phase 2 CRO run. At current prices, the 140g needed costs approximately \$12,600–\$21,000. If rhenium prices double again to \$12,000/kg, the precursor alone would cost \$36,000.

Mitigation Action

Priority

Saving Potential

Fix-price CRO contract locking in $[\text{Re}(\text{CO})_5\text{Br}]$ cost now

Priority 1 — act immediately

Avoids unknown upside cost

Bulk-buy $[\text{Re}(\text{CO})_5\text{Br}]$ (500g–1kg) upfront as 12-month buffer

Priority 2

\$20k–\$40k saved if prices double

CRO to synthesise $[\text{Re}(\text{CO})_5\text{Br}]$ from $\text{Re}_2(\text{CO})_{10} + \text{Br}_2$ directly

Priority 3

~30% reduction in precursor cost

Monitor US USGS critical mineral re-classification

Ongoing watch

Export control risk if Re designated strategic

6. Phase 3 — Pilot Scale: 5,000 Sensors at Low-Cost Nodes

6.1 The Phase 3 Premise

Phase 3 is a pilot production run of 5,000 sensors, triggered by successful validation of Phase 1 use cases. It rests on three structural changes from Phase 1:

First, all Re sensitizer is supplied centrally from the Phase 2 CRO run. No site in Phase 3 performs any Batch 1 synthesis. The glovebox requirement disappears entirely from Phase 3. Every site in Phase 3 only needs to do Batch 2: SWCNT-TiO₂ preparation, spray-coating, and sensitizer soaking.

Second, fabrication is concentrated exclusively at low-cost nodes. India (IIT Bombay, IIT Madras, IISc Bangalore) and South Korea (KAIST, SNU, POSTECH) are the primary Phase 3 sites, with Indian labs taking 60% of volume and Korean labs taking 40%. The 15-fold difference in effective hourly rate between Switzerland and India means that concentrating Batch 2 at Indian and Korean nodes produces dramatic per-sensor cost reduction.

Third, materials are procured in bulk. Electrode unit prices at 10,000+ units are approximately 50–65% below the retail price of a pack of 50. SWCNT bulk pricing at 20g scale is approximately 70% below per-gram research pricing.

6.2 Phase 3 Full Cost Model — 5,000 Sensors

Total Phase 3 cost ~\$370,000 <i>materials + labour + logistics</i>	Per sensor cost ~\$74 <i>97% reduction vs Phase 1</i>	Material cost ~\$265,000 <i>electrodes dominate at 56%</i>
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Cost Item

Basis

Cost

IDEAU200 gold IDE electrodes ×6,250 (25% yield buffer)

Bulk OEM pricing ~\$3.50–\$5.00/unit

~\$22,000–\$31,000

[(Pbpy)(CO)₃ReBr] — amortised from Phase 2 CRO run

100g covers all of Phase 3 with large reserve

~\$42,000 (full Phase 2 cost)

P3-SWNT (~1g total at 0.2mg/sensor)

Bulk pricing ~\$150/g at this quantity

~\$150

Ti(OiPr)₄ — small quantity, bulk drum shared

Fractional use of 5L drum

~\$200

Anhydrous DMSO, EtOH, solvents — bulk industrial grade

Bulk drums at industrial pricing

~\$3,000

Nylon membranes, crimp vials, airbrush consumables, Ar gas

100 fabrication batches × 50 sensors

~\$4,000

QC: resistance testing, spot-check UV-vis, 5% sampling

In-lab, low marginal cost

~\$1,500

Cold-chain logistics: Re complex to nodes, sensors outbound

International freight, temperature controlled

~\$4,500

Labour — India node (60%): 120 person-days at ~\$45/day

IIT/IISc postdoc effective rate

~\$5,400

Labour — Korea node (40%): 80 person-days at ~\$72/day

KAIST/SNU postdoc effective rate

~\$5,760

QC coordination overhead (20% on all labour)

Supervision, protocol adherence

~\$2,230

TOTAL

~\$90,000–\$99,000*

* Note: The Re sensitizer line item of ~\$42,000 represents the full cost of the Phase 2 CRO run. Amortised across Phase 3 only, it amounts to ~\$8.40/sensor. If Phase 2 is treated as a sunk cost prior to Phase 3 launch, the incremental Phase 3 cost is approximately \$48,000–\$57,000, or ~\$9.60–\$11.40/sensor. The full economic model shown above (\$90k–\$99k) includes the Phase 2 Re synthesis cost in full to give a complete view of total programme investment.

6.3 The Electrode Cost Lever

The electrode substrate is 50%+ of Phase 3 material cost. Replacing branded DropSens IDEAU200 with a validated custom OEM equivalent is the single highest-leverage cost reduction available in the entire programme.

Electrode Option

Unit Price

Total for 6,250 units

Validation Required

DropSens IDEAU200 (branded, retail)

~\$10/unit

~\$62,500

None — reference standard

DropSens IDEAU200 (bulk OEM, 10k+ order)

~\$3.50–\$5.00/unit

~\$22,000–\$31,000

Confirm pricing with Metrohm

Chinese OEM custom IDE (Al₂O₃ ceramic, gold, 100µm pitch)

~\$0.80–\$1.50/unit

~\$5,000–\$9,375

200-unit geometry validation experiment required

A 200-unit geometry validation experiment costs approximately \$3,000 and should be run in parallel during Phase 1. If the custom OEM electrode is confirmed equivalent, the Phase 3 electrode cost is reduced by \$17,000–\$26,000, and the per-sensor cost falls to approximately \$65–70. This experiment is strongly recommended before placing Phase 3 electrode procurement.

6.4 Phase 3 Labour Model

Node

Volume

Fabrication Batches

Person-Days

Labour Cost

India (IIT Bombay, IIT Madras, IISc)

3,000 sensors (60%)

60 batches × 50 sensors

120 days

~\$5,400

South Korea (KAIST, SNU, POSTECH)

2,000 sensors (40%)

40 batches × 50 sensors

80 days

~\$5,760

QC coordination overhead (20%)

—

—

40 days effective

~\$2,230

TOTAL

5,000 sensors

100 batches

240 person-days

~\$13,390

7. Use Case Validation and the Phase 4 Decision Gate

7.1 What Validation Must Confirm

The decision to proceed from Phase 3 (5,000 sensors) to Phase 4 (100,000 sensors) is gated on use-case validation, not just protocol transfer. The sensor is rated at Technology Readiness Level 5 (TRL 5) by ETH Transfer (Reference 2024-034): a prototype has been built and tested in the lab, with a patent pending. What must now be confirmed is whether it performs well enough in real deployment environments to justify a \$1.4M production commitment. The ETH

transfer sheet and the published paper together define seven specific target application domains, each with distinct validation requirements.

The ETH transfer sheet (Reference 2024-034) and the published paper together identify seven specific target application domains. These are the segments against which Phase 3 sensors must be validated:

Application Domain

Target O₂ Range

Key Challenge to Validate

Success Threshold

Exhaust gas analysis — automobile and industrial process control

ppm-% O₂ (exhaust: 0.1–15%)

Confirmed selectivity vs. CO, CO₂, CH₄, C₂H₄ (paper Fig. 5A). Validate at elevated temperature and soot/particulate exposure in real exhaust stream.

No cross-response to CO/CO₂/hydrocarbons; stable at 50–200°C

Water and air quality monitoring

0.1 ppm – 21% O₂

Dissolved O₂ equivalent requires aqueous-phase adaptation. Air quality: confirmed humidity tolerance 0–80% RH (paper Fig. S32). Low-power field deployment <0.6 W (ETH transfer sheet).

Stable >6 months outdoor; operation at 0–80% RH confirmed

Early detection of food spoilage (modified atmosphere packaging)

100 ppm – 2% O₂ (headspace)

Must operate through or integrated into packaging; temperature range 2–25°C. Sensitivity at low ppm levels confirmed (LOD 2.2 ppm at 1 min exposure, paper Fig. 4).

LOD ≤5 ppm; stable at 2–25°C; no response to CO₂, C₂H₄ (spoilage gases)

Breath monitoring and personal health

15–21% O₂ (exhaled breath: ~16%)

Sensor operates under ambient O₂ with confirmed sensitivity to concentration changes of ≥26 ppm (paper Fig. 6). Miniature design and <0.6 W power confirmed. Biocompatibility not yet assessed.

Detects ≥ 26 ppm ΔO_2 change under ambient air; miniature form factor; < 0.6 W
 Personnel health protection (confined spaces, industrial O_2 depletion alarm)
 16–21% O_2 (depletion alert at $< 18\%$)
 Sensitivity to O_2 concentration decreases confirmed under ambient air carrier gas (paper Fig. 6).
 Rapid $t_{90} = 6$ s response is suitable for alarm applications. Must validate battery-powered operation.
 Response < 10 s to 1% O_2 drop; stable > 3 months; operable from small battery
 Food and pharmaceutical process control (inert gas purity, N_2 blanket)
 20–500 ppm O_2 (trace monitoring)
 Confirmed selectivity over N_2O , CO_2 , H_2 , CH_4 , N_2 (paper Fig. 5A). TRL 5 as per ETH transfer sheet — prototype built and lab-tested. Must validate against regulatory standards (e.g. pharmacopoeia O_2 limits).
 LOD < 200 ppb (paper S30); no cross-response; stable > 3 months ambient
 Cell health and disease progression monitoring (biomedical research)
 1–200 ppm O_2 (hypoxia: < 20 ppm dissolved equivalent)
 Paper cites cell health monitoring as target application. Aqueous-phase adaptation required.
 ppb-level LOD (157 ppb at 60 min) relevant for hypoxia detection. Biocompatibility study required.
 LOD 157 ppb at 60 min exposure; selectivity vs. biological interferants to be assessed

7.2 The 4-Week Aging Observation

A key finding from the paper: SWCNT-TiO₂-Re electrodes improve significantly after 4 weeks of aging in water dispersion prior to sensitization. Ostwald ripening increases TiO₂ surface area and dye loading. This 4-week wait is non-negotiable and must be built into all Phase 3 and Phase 4 production schedules.

7.3 Phase 4 Decision Gate Criteria

Phase 4 (100,000 sensors) should be authorised when all of the following are confirmed from Phase 3 data:

- At least one use case with a confirmed commercial partner and purchase intent for a minimum of 50,000 sensors/year
- Field performance matching laboratory specifications to within 20% across at least two independent deployment sites
- Six-month field stability data (at least 80% of Phase 3 sensors still within spec at month 6)
- Custom OEM electrode validated as equivalent to DropSens IDEAU200
- Phase 2 CRO has demonstrated capacity and willingness to supply at Phase 4 volume (multiple 100g runs per year)
- Rhenium supply secured via fixed-price contract covering at least 18 months of Phase 4 production

8. Phase 4 — Full Scale: 100,000 Sensors per Year

8.1 Phase 4 Cost Model

Phase 4 assumes all Phase 3 validations have passed: OEM electrode is confirmed, Re supply is secured at fixed price, and at least one commercial partner with high-volume purchase intent is in place. Fabrication remains at India and Korea nodes, with the possibility of adding one additional low-cost node (e.g. UNICAMP Brazil or SKKU Korea) to handle volume and provide redundancy.

Phase 4 total cost	Per sensor (OEM electrode)	Per sensor (DropSens bulk)
~\$1.38M	~\$9.50	~\$13.80

100,000 sensors/year

if custom IDE validated

if branded electrode retained

Cost Item

Basis

Phase 4 Cost

Gold IDE electrodes ×125,000 (20% yield buffer)

Custom OEM ~\$1.20/unit OR bulk DropSens ~\$4.00/unit

\$150,000 OR \$500,000

[(Pbpy)(CO)₃ReBr] — multiple CRO batches

~3 × 100g runs/year at \$42k each

~\$126,000/year

P3-SWNT ~20g total (bulk)

~\$150/g at 20g scale

~\$3,000

Ti(OiPr)₄, solvents, bulk industrial grade

5L drum + bulk DMSO/EtOH

~\$13,600

Consumables (membranes, vials, Ar, airbrush)

2,000 batches × 50 sensors

~\$48,000

QC sampling protocol (5%)

Resistance + spot-check UV-vis per 20 batches

~\$15,000

Cold-chain logistics (Re in, sensors out)

Annual programme

~\$18,000

Labour — India (60%): 2,400 person-days at \$45/day

IIT/IISc postdoc effective rate

~\$108,000

Labour — Korea (40%): 1,600 person-days at \$72/day

KAIST/SNU postdoc effective rate

~\$115,200

QC coordination (20% on labour)

~\$44,640

TOTAL (OEM electrode) / (DropSens bulk)

\$641,240 / \$991,240

8.2 Phase 4 Scale Economics Summary

Scenario

Volume

Per-Sensor Cost

Key Assumption

Phase 1 (university distributed)

100 sensors

~\$680/sensor

20× parallel Batch 1 synthesis

Phase 3 pilot (CRO Re supply)

5,000 sensors

~\$74/sensor (incl. Phase 2)

Low-cost nodes only, no glovebox in Batch 2

Phase 3 incremental only

5,000 sensors

~\$10–\$12/sensor

Phase 2 Re synthesis treated as sunk cost

Phase 4 — OEM electrode validated

100,000 sensors

~\$9.50/sensor

Custom IDE + CRO Re at \$42k/batch

Phase 4 — DropSens bulk pricing

100,000 sensors

~\$13.80/sensor

Branded electrode retained at bulk OEM price

9. Risk Register

Risk**Probability****Impact****Mitigation**

Re metal price continues rising beyond \$10,000/kg

Medium

High — adds \$20k–\$40k to CRO run

Fix-price CRO contract; bulk buy $[\text{Re}(\text{CO})_5\text{Br}]$ now

US/USGS export controls on Re as strategic mineral

Low-Medium

Very High — supply chain disruption

Source $[\text{Re}(\text{CO})_5\text{Br}]$ from non-US CRO (Germany, India)

Protocol fails to transfer at some Phase 1 sites

Medium

Low — affects <20 sensors

Peer-to-peer protocol sharing; ETH Zürich advisory

OEM electrode fails geometry validation

Low-Medium

Medium — raises Phase 3/4 cost

Validate in Phase 1 before placing Phase 3 order

4-week aging step forgotten at Phase 3 nodes

Medium

High — batch of sensors underperforms

Written SOP, mandatory; QC check UV-vis λ_{MLCT} before soaking

SWCNT batch variability (P3-SWNT lot-to-lot)

Low

Medium — resistance variation in spray-coating

Buy 1-year supply in single lot; lot qualification test

Use case fails field validation — Phase 4 not triggered

Medium

High — strands 5,000 sensors and Phase 2 investment

Phase 3 sensors are functional devices with real LOD 157 ppb; multiple use cases reduce single-point dependency

Competing commercial sensor enters market pre-Phase 4

Medium

Medium — price pressure

Visible-light operation and ppb-range LOD are differentiating; patent analysis recommended

**10. Integrated
Programme
Roadmap****Phase****Activity****Timeline**

Cost

Gate to Next Phase

Phase 1

100 sensors across 20 universities; protocol transfer; use-case testing; OEM electrode validation

Months 1–12

~\$68,000

Protocol transfer at $\geq 15/20$ sites; use-case data from ≥ 2 applications

Phase 2

CRO contract for 100g [(Pbpy)(CO)₃ReBr]; fix-price Re metal; CoA delivery

Months 4–8 (parallel)

~\$42,000

Full CoA: NMR, IR, UV-vis, HPLC; argon-sealed delivery to nodes

Phase 3

5,000 sensors at India + Korea nodes; Batch 2 only; centralised Re supply

Months 9–18

~\$90,000–\$99,000

6-month field stability; commercial partner purchase intent; OEM electrode decision

Phase 4

100,000 sensors/year; full programme; OEM electrode if validated

Months 18+

~\$641k–\$991k/year

Ongoing commercial programme

Total programme investment to Phase 4 entry: ~\$200,000–\$210,000 (Phases 1+2+3 combined). This is the validation budget required before committing to \$641k–\$991k/year in volume production.

Appendix A: Complete Material and Equipment Lists

**A.1 Batch 1 —
Re Sensitizer
Synthesis (per**

run)

Material

Sigma-Aldrich / Supplier Cat. No.

Unit Price

Amount per Run

Ti(OiPr)₄ 99.999%

Sigma 377996

~\$80/25mL

50 µL

[Re(CO)₅Br]

Sigma 327638

~\$150/1g

1.0 eq (~1g)

4,4'-Dibromo-2,2'-bipyridine

Chem-Impex / abcr

~\$133/1g

1.0 eq

Pd(PPh₃)₄ 99%

Sigma 216666

~\$95/1g

0.05 eq

PPh₃ 99%

Sigma T84409

~\$35/100g

5.0 eq

Diethyl phosphite

Sigma

~\$30/100mL

2.3 eq

Bromotrimethylsilane 97%

Sigma 194409

~\$45/25mL

68 eq

Et₃N

Sigma

~\$30/100mL

2.3 eq

Anhydrous toluene Sure/Seal

Sigma 244511

~\$40/100mL

60 mL

Anhydrous CHCl₃ Sure/Seal

Sigma 288306

~\$40/100mL

As needed

Anhydrous MeOH Sure/Seal

Sigma 322415

~\$35/100mL

3 mL + washes
Silica gel for chromatography
Sigma
~\$30/500g
Column prep

**A.2 Batch 2 —
Electrode
Fabrication (per 50-
electrode run)**

Material	Supplier	Unit Price	Amount per Run
IDEAU200 gold IDE electrodes			
Metrohm DropSens DRP-IDEAU200-U50			
		~\$10/unit	
62 units (25% yield buffer)			
P3-SWNT >90%			
Carbon Solutions Inc.			
		~\$280/g (research); ~\$150/g (bulk)	
		~0.01g (10 mg)	
Ti(OiPr) ₄ 99.999%			
Sigma 377996			
		~\$80/25mL	
50 µL			
Anhydrous DMSO Sure/Seal			
Sigma 276855			
		~\$45/100mL	
2 mL per electrode soak			
Anhydrous EtOH Sure/Seal			
Sigma 459836			
		~\$35/100mL	
Dispersion and washes			
[(Pbpy)(CO) ₃ ReBr] (Phase 2 supply)			
CRO Phase 2			
		~\$420/g (CRO)	

~0.5 mL of 0.2 mM DMSO
Nylon membrane filters 0.2 μ m
Sigma / Merck
~\$0.50/filter
1 per run
Crimp vials, Ar gas, consumables
Lab supply
~\$30–40/run
—

A.3 Equipment List — Tier Requirements

Equipment
Required Tier
Estimated Cost
Notes

Argon-filled glovebox
Tier 1 only
\$30,000–\$80,000
Required for Batch 1 complexation step
Schlenk line / vacuum manifold
Tier 1 only
\$2,000–\$5,000
Alternative to glovebox for experienced operators
Microwave reactor (Biotage Initiator+)
Tier 1 (or oil bath)
\$20,000–\$40,000

Conventional oil bath acceptable for Step 2a
Rotary evaporator
Tier 1 only
\$3,000–\$8,000
Required for solvent removal between steps
Flash chromatography (Biotage Isolera)
Tier 1 only
\$15,000–\$30,000
Required for Step 1 purification
High vacuum pump
Tier 1 only
\$1,500–\$5,000
Required for product drying (overnight under HV)
NMR spectrometer 300–500 MHz
Tier 1 (core facility)
\$300k–\$800k or facility
 $^1\text{H} + ^{31}\text{P}$ required for batch confirmation
FT-IR spectrometer
Tier 1 (in-lab)
\$15,000–\$40,000
CO bands at 2033 and 1915 cm^{-1} required
Airbrush (Iwata Revolution BR)
Tier 2
~\$80
Batch 2 spray-coating — no special facility needed
Heating plate (150°C capable)
Tier 2
\$300–\$800
Batch 2 spray substrate temperature
Syringe pump
Tier 2
\$500–\$1,500
Controlled water addition to SWCNT-TiO₂ mixture
Ultrasonic bath
Tier 2
\$500–\$3,000
SWCNT-TiO₂ dispersion preparation
Agilent DAQ970A + multiplexer
Tier 2
\$3,000–\$6,000
Resistance measurement of finished electrodes
Green LED (LCFOCUS LC-10RGB-C30)
Tier 2
~\$50
Sensing activation (200 mW/mm² at 2cm)

Appendix B: Sensor Performance Specifications

Parameter	Value	Notes
Limit of detection (LOD)	157 ppb O ₂	Under green LED illumination, ambient conditions
Detection range	20 ppm – 200,000 ppm (20%)	Linear range in chemiresistive response
Response time (t ₉₀)	~6 seconds	To 1,000 ppm O ₂ under green LED
Recovery time	~9 minutes	After 2 min exposure to 1,000 ppm O ₂
Storage stability	>3 months	Dark storage, ambient conditions
Operational cycling stability	>6 hours continuous	Repeated exposure/recovery cycles
Humidity tolerance	0–80% RH	No measurable performance degradation
Selectivity	No response to N ₂ O, CO ₂ , H ₂ , CH ₄ , C ₂ H ₄ , CO, N ₂	Unique to O ₂ via photochemical mechanism
Target device resistance	1–5 kΩ	

QC criterion for spray-coated Batch 2 electrodes

Sensitizer concentration (soaking)

0.2 mM in DMSO

Optimal from screening; 16h dark soak

LED parameters

~200 mW/mm² green, 0.08A, 2.0 cm distance

LCFOCUS LC-10RGB-C30

TiO₂ phase

Amorphous sol-gel

Outperforms anatase, P25, rutile at same loading

Excited-state potential E(S^{*}/S⁺)

−0.93 V vs NHE

Thermodynamic driver for O₂ reduction

End of Document

SWCNT–TiO₂–Re Sensor: From Supplementary Data to Distributed Manufacturing

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